Electric Flux Density

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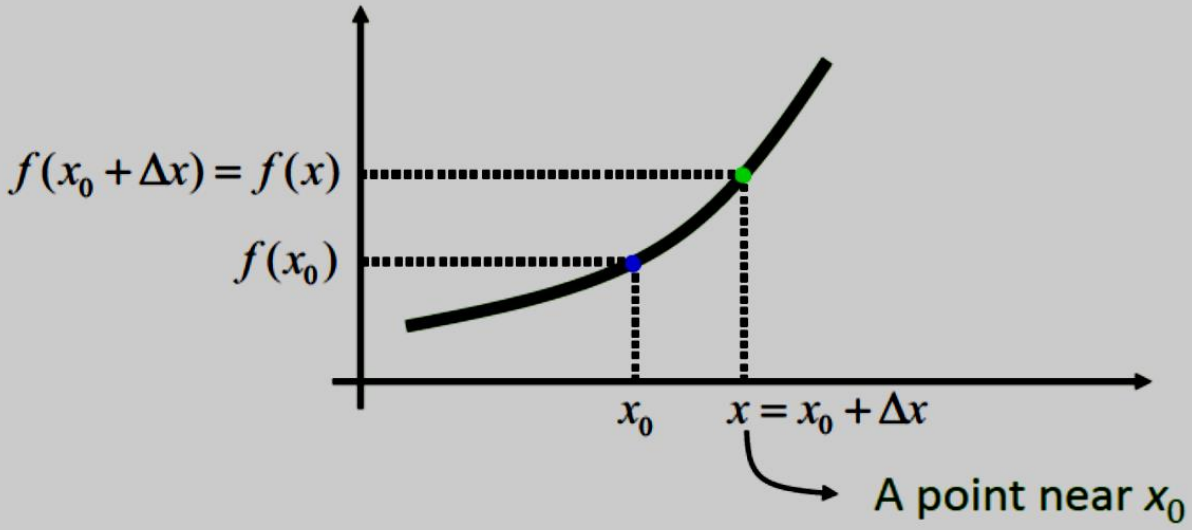
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Application of Gauss's Law: Differential volume element

- We are now going to apply the methods of Gauss's law to a slightly different type of problem: a surface without symmetry.
- We have to choose such a very small closed surface that D is almost constant over the surface, and the small change in D may be adequately represented by using the first two terms of the Taylor's-series expansion for D.
- The result will become more nearly correct as the volume enclosed by the Gaussian surface decreases.
- We intend eventually to allow this volume to approach zero.

Taylor's Series Expansion



$$f(x)=f(x_0+\Delta x)$$

$$f(x)=f(x_0)+\frac{f'(x_0)}{1!}\Delta x+\frac{f''(x_0)}{2!}(\Delta x)^2+\cdots+\frac{f^n(x_0)}{n!}(\Delta x)^n$$

Only the linear terms are used for the linearization

Application of Gauss's Law: Differential volume element

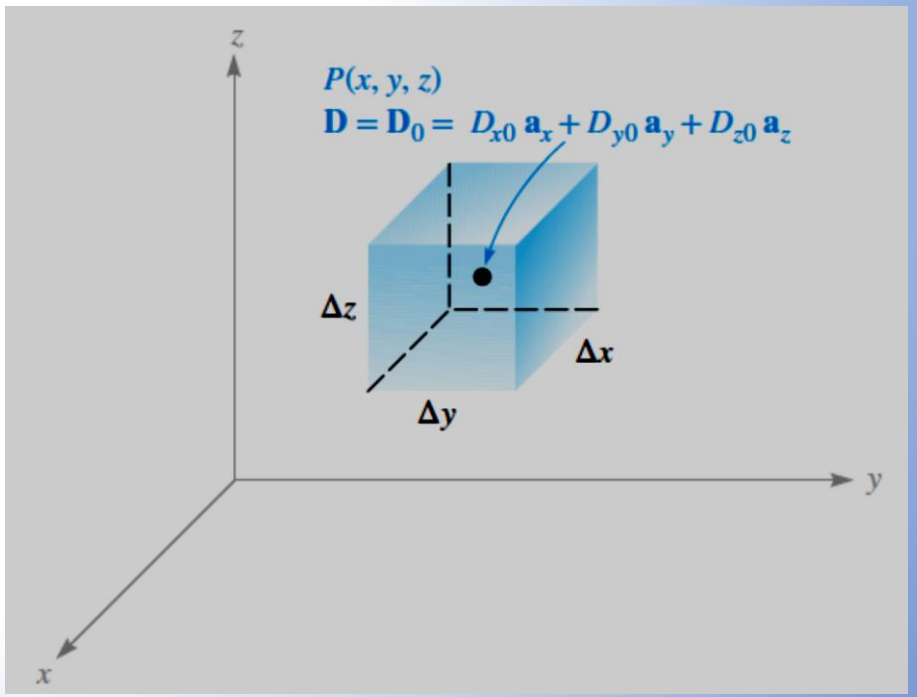
- Consider any point P, located by a rectangular coordinate system.
- The value of D at the point P may be expressed in rectangular components:

$$\mathbf{D}_0 = D_{x0}\mathbf{a}_x + D_{y0}\mathbf{a}_y + D_{z0}\mathbf{a}_z$$

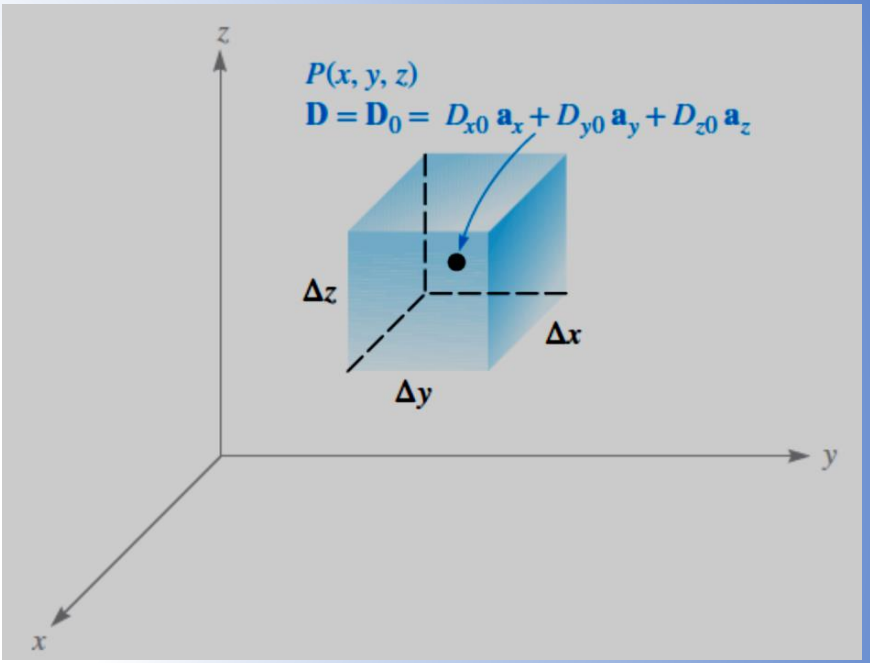
- We now choose as our closed surface, the small rectangular box, centered at P, having sides of lengths Δx, Δy, and Δz, and apply Gauss's law:

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = Q$$

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{front}} + \int_{\text{back}} + \int_{\text{left}} + \int_{\text{right}} + \int_{\text{top}} + \int_{\text{bottom}}$$



Application of Gauss's Law: Differential volume element



- We will now consider the front surface in detail.
- The surface element is very small, thus D is essentially constant over this surface:

$$\begin{aligned}\int_{\text{front}} &= \mathbf{D}_{\text{front}} \cdot \Delta\mathbf{S}_{\text{front}} \\ &= \mathbf{D}_{\text{front}} \cdot \Delta y \Delta z \, \mathbf{a}_x \\ &= D_{x,\text{front}} \Delta y \Delta z\end{aligned}$$

- The front face is at a distance of $\Delta x/2$ from *P*, and therefore:

$$\begin{aligned}D_{x,\text{front}} &= D_{x0} + \frac{\Delta x}{2} \times \text{rate of change of } D_x \text{ with } x \\ &= D_{x0} + \frac{\Delta x}{2}\frac{\partial D_x}{\partial x}\end{aligned}$$

Application of Gauss's Law: Differential volume element

- We have now, for front surface:

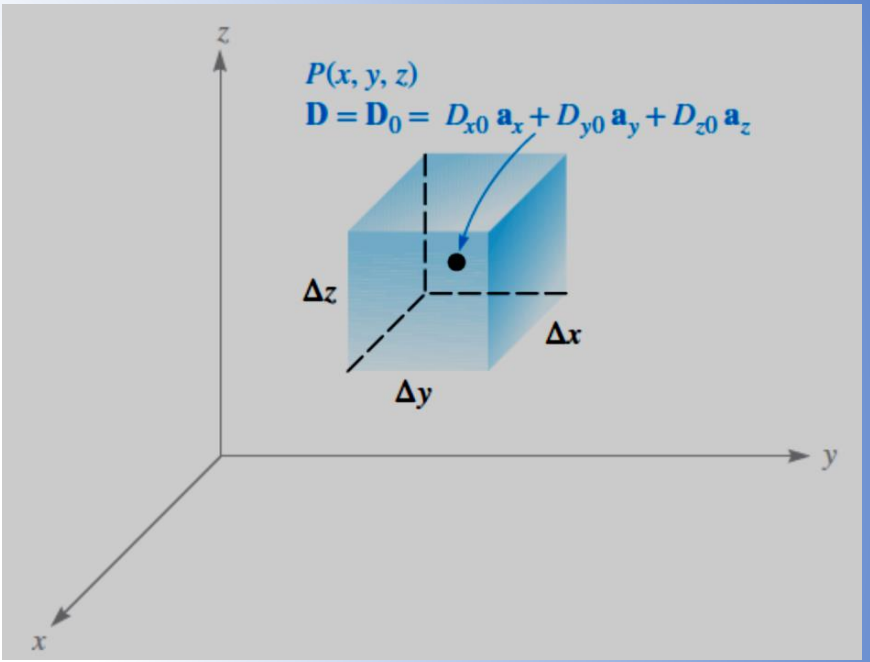
$$\int_{\text{front}} = \left(D_{x0} + \frac{\Delta x}{2}\frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

- In the same way, the integral over the back surface can be found as:

$$\int_{\text{back}} = \mathbf{D}_{\text{back}} \cdot \Delta \mathbf{S}_{\text{back}}$$
$$= \mathbf{D}_{\text{back}} \cdot (-\Delta y \Delta z \ \mathbf{a}_x)$$
$$= -D_{x,\text{back}} \Delta y \Delta z$$

$$D_{x,\text{back}} = D_{x0} - \frac{\Delta x}{2}\frac{\partial D_x}{\partial x}$$

$$\int_{\text{back}} = \left(-D_{x0} + \frac{\Delta x}{2}\frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$



Application of Gauss's Law: Differential volume element

- If we combine the two integrals over the front and back surface, we have:

$$\int_{front} + \int_{back} = \frac{\partial D_x}{\partial x}\Delta x\Delta y\Delta z$$

- Repeating the same process to the remaining surfaces, we find:

$$\int_{right} + \int_{left} = \frac{\partial D_y}{\partial y}\Delta y\Delta x\Delta z$$

$$\int_{top} + \int_{bottom} = \frac{\partial D_z}{\partial z}\Delta z\Delta x\Delta y$$

- These results may be collected to yield:

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Delta x\Delta y\Delta z$$

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = Q = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Delta v$$

■ Example

Let $D = y^2z^3\,a_x + 2xyz^3\,a_y + 3xy^2z^2\,a_z$ nC/m² in free space. (*a*) Find the total electric flux passing through the surface $x = 3$, $0 \le y \le 2$, $0 \le z \le 1$ in a direction away from the origin. (*b*) Find |E| at *P*(3,2,1). (*c*) Find an approximate value for the total charge contained in an incremental sphere having a radius of 2 mm centered at *P*(3,2,1).

(a)
$$\psi = \int_S \mathbf{D}_S \cdot d\mathbf{S}$$
$$= \int_{z=0}^{1}\int_{y=0}^{2} \left(y^2z^3\mathbf{a}_x + 2xyz^3\mathbf{a}_y + 3xy^2z^2\mathbf{a}_z\right)\cdot\left(dydz\ \mathbf{a}_x\right)\Big|_{x=3}$$
$$= \int_0^1\int_0^2 y^2z^3\,dydz$$
$$= \tfrac{1}{3}y^3\Big|_0^2\ \tfrac{1}{4}z^4\Big|_0^1$$
$$= \underline{\tfrac{2}{3}\text{ nC}}$$

(b)
$$\mathbf{D} = y^2z^3\mathbf{a}_x + 2xyz^3\mathbf{a}_y + 3xy^2z^2\mathbf{a}_z$$
$$\mathbf{D}_P = (2)^2(1)^3\mathbf{a}_x + 2(3)(2)(1)^3\mathbf{a}_y + 3(3)(2)^2(1)^2\mathbf{a}_z$$
$$= 4\mathbf{a}_x + 12\mathbf{a}_y + 36\mathbf{a}_z\ \text{nC/m}^2$$

$$|\mathbf{D}_P| = D_P = \sqrt{(4)^2 + (12)^2 + (36)^2}$$
$$= 38.158\,\text{nC/m}^2$$

$$|\mathbf{E}_P| = \frac{|\mathbf{D}_P|}{\varepsilon_0}$$
$$= \frac{38.158\ \text{nC/m}^2}{8.854\times10^{-12}}$$
$$= \underline{4.31\ \text{kV/m}}$$

(c)
$$Q = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Delta v$$
$$Q_P = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Big|_P \Delta v$$
$$= \left(0 + 2xz^3 + 6xy^2z\right)\Big|_{\substack{x=3\\y=2\\z=1}} \text{nC/m}^3 \times \tfrac{4}{3}\pi(2\times10^{-3})^3\,\text{m}^3$$
$$= \left(0 + 2(3)(1)^3 + 6(3)(2)^2(1)\right)\times\tfrac{4}{3}\pi(2\times10^{-3})^3\,\text{nC}$$
$$= \underline{2.61\times10^{-15}\ \text{C}}$$


Thank
you